Benton County - Vertical Equity Deep Dive

2010-2025 using time-adjusted sales (ref=2026-01-01)

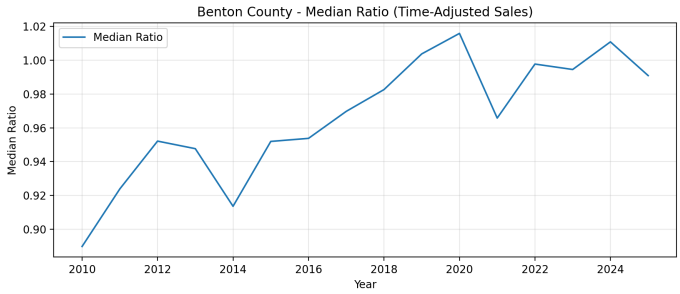
Key finding: a measurable structural improvement begins in 2016 for vertical equity (PRD).
Best breakpoints (simple two-regime fit): PRD 2015-2016, COD 2017-2018, PRB slope 2017-2018.
Recent performance: 2023-2024 show near-neutral PRD and strong uniformity; 2025 shows mild upward drift to monitor.

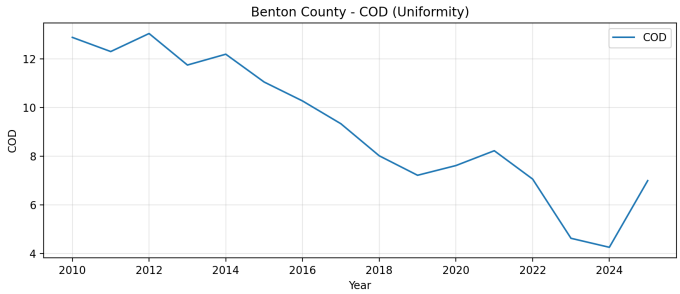
What this report contains

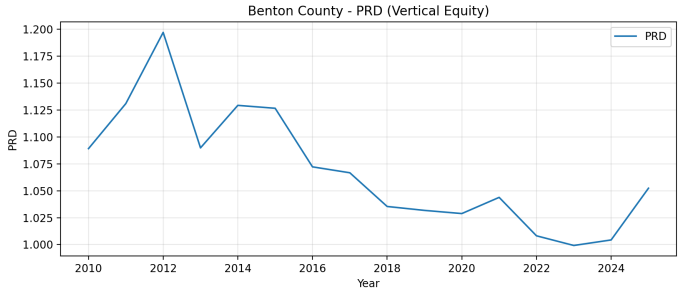
- Annual ratio study metrics: Median Ratio, COD (uniformity), PRD and PRB slope (vertical equity).
- Estimated countywide price index (from median sale prices) used to time-adjust sales.
- Neighborhood (hood) diagnostics: where PRD improved most after 2016 and where it worsened.
- Action plan: how Benton can lock in #1-tier vertical equity performance and become a replicable standard.

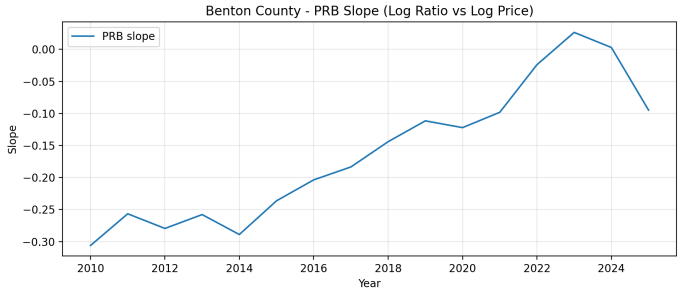
Annual Metrics (2010-2025)

Median Ratio, COD, PRD, PRB slope



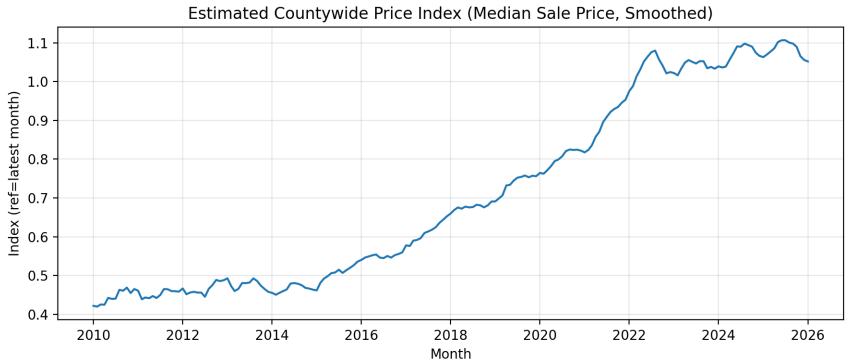


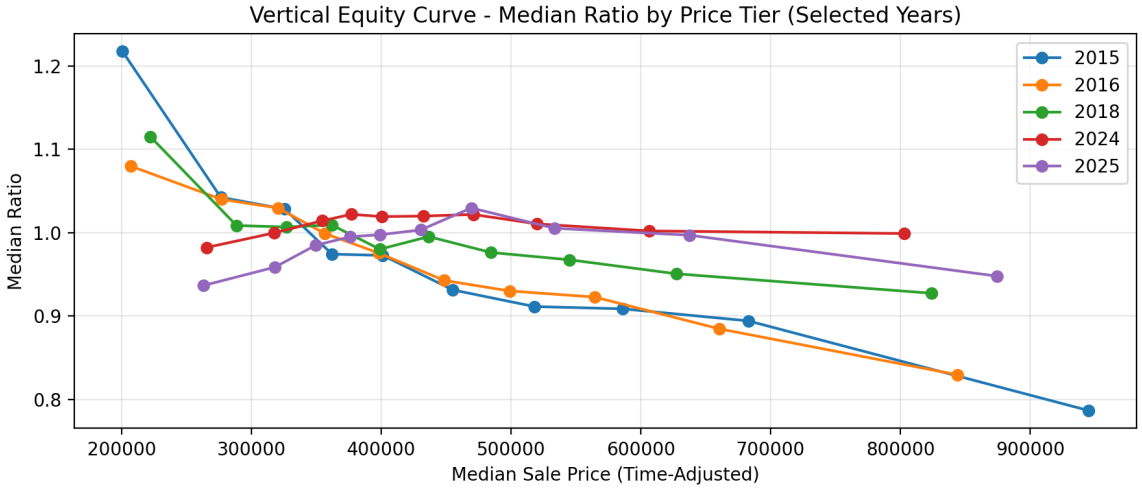




Price Index & Tier Curves

Time adjustment and vertical equity curves

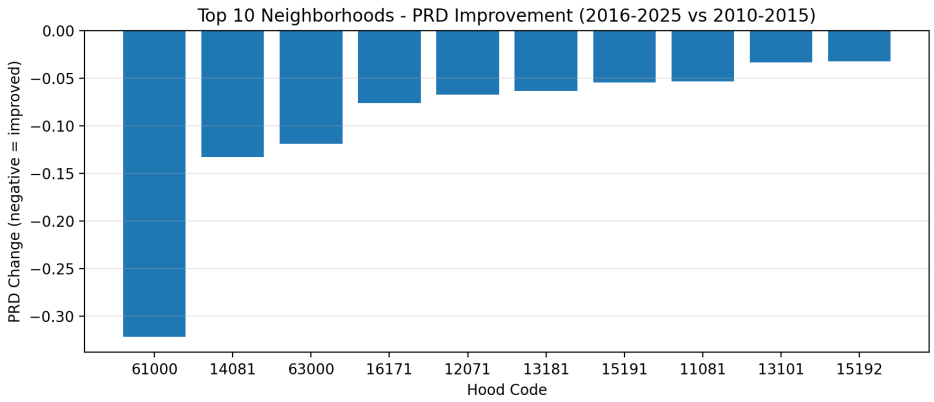


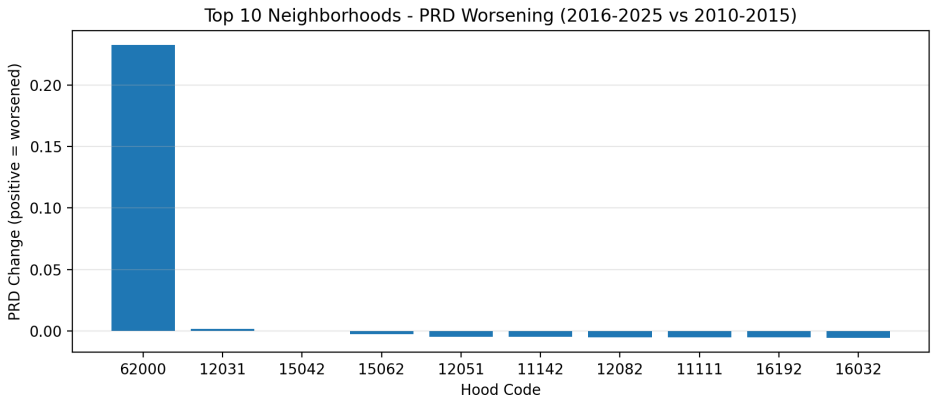


Interpretation: flatter tier curves with ratios near 1.00 across price levels indicates stronger vertical equity.

Neighborhood Diagnostics (Hood Codes)

PRD change: 2016-2025 vs 2010-2015





Watchlist

Hood 62000: PRD 0.955 -> 1.187 (Δ +0.232), n=251
Hood 12031: PRD 1.011 -> 1.013 (Δ +0.002), n=344
Hood 15042: PRD 1.005 -> 1.005 (Δ +0.000), n=179
Hood 15062: PRD 1.005 -> 1.002 (Δ -0.003), n=411

How Benton Becomes the Standard

Operational playbook to sustain top-tier vertical equity

1) Lock-in the 2016 structural gains

- Treat 2016 as the point where vertical equity measurably improved (PRD break). Preserve the practices that produced it: model governance, consistent sales validation, and systematic recalibration.
- Make PRD and PRB slope first-class acceptance tests for every neighborhood adjustment and every annual table roll.

2) Add a 'vertical equity gate' to every calibration cycle

- Before/after dashboards must show: (a) tier curves, (b) PRD, (c) PRB slope, (d) ratio distribution by value decile, and (e) a hood-level watchlist. No roll to production without passing thresholds.
- Target thresholds (adapt to local standards): PRD near 1.00 and PRB slope near 0.00, with stability across tiers and neighborhoods.

3) Focus resources where vertical equity is still brittle

- Use the hood diagnostics: the 'worst PRD change' list is where drift is most likely. Prioritize data cleanup, feature auditing, and stratification refinement in those hoods first.
- Instrument the pipeline: if 2025-like drift appears, trigger a targeted review (sales mix, new construction bias, land-only leakage, and model extrapolation issues).

4) Publish a replicable standard

- Package Benton Method governance: data hygiene -> sales validation -> model calibration -> vertical equity gate -> taxpayer-facing explainers. Make it open, auditable, and repeatable.
- Use the same exhibits year over year so partners can see trend continuity and trust the process.